Example 2.19 A picture of length 23 *cm* and breadth 11 *cm* is painted on a chart, such that there is a margin of 3 *cm* along each of its sides. Find the total area of the margin.

Solution

Here $L = 23\ cm\ B = 11\ cm$

$$\text{Area of the chart} = L \times B$$
$$= 23 \times 11$$
$$= 253\ cm^2$$

$l = L - 2w = 23 - 2(3) = 23 - 6 = 17\ cm$

$b = B - 2w = 11 - 2(3) = 11 - 6 = 5\ cm$

Area of the picture $17 \times 5 = 85\ cm^2$

Therefore, the area of the margin $= 253 - 85$

$$= 168\ cm^2.$$

Example 2.20 A verandah of width 3 *m* is constructed along the outside of a room of length 9 *m* and width 7 *m*. Find (a) the area of the verandah (b) the cost of cementing the floor of the verandah at the rate of ₹15 per *sq.m.*

Solution

Here, $l = 9\ m,\quad b = 7\ m$

$$\text{Area of the Room} = l \times b$$
$$= 9 \times 7$$
$$= 63\ m^2$$
$$L = l + 2w = 8 + 2(3) = 8 + 6 = 14\ m$$
$$B = b + 2w = 5 + 2(3) = 5 + 6 = 11\ m$$

Area of the room including verandah $= L \times B$

$$= 14 \times 11$$
$$= 154\ m^2$$

The area of the verandah = Area of the room including verandah − Area of the room

$$= 154 - 63$$
$$= 91\ m^2$$

The cost of cementing the floor for $1 sq.m$ = ₹15

Therfore, the cost of cementing the floor of the verandah $= 91 \times 15 = ₹1365.$

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